

Natural Language Processing Project

AI4303 – NLP Course

Sentiment Classification with Hugging Face

Section 2

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Team Members

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Introduction

Natural Language Processing (NLP) is an important field in Artificial Intelligence that focuses on enabling computers to understand and analyze human language. One common NLP task is sentiment analysis, which determines whether a text expresses positive or negative sentiment.

In this project, a sentiment classification system was developed using three pretrained Hugging Face transformer models: DistilBERT, RoBERTa, and multilingual BERT. The models were evaluated using the IMDb movie review dataset.

Several evaluation metrics were used, including Accuracy, Precision, Recall, and F1-score. In addition, a Streamlit web application was developed to allow users to test custom sentences and receive sentiment predictions with confidence scores.

Dataset Description

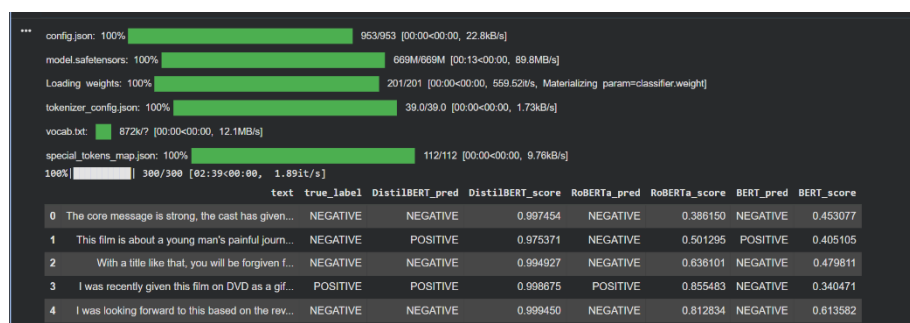
The IMDb movie review dataset from the Hugging Face datasets library was used in this project. A balanced subset of 300 reviews was selected, including:

- 150 positive reviews
- 150 negative reviews

The dataset was converted into a pandas DataFrame and preprocessed before evaluation. The preprocessing steps included:

1. Selecting the required columns
2. Renaming the label column to true_label
3. Converting labels into:
 - POSITIVE
 - NEGATIVE
4. Shuffling the dataset randomly

The final cleaned dataset was saved into Excel and CSV files for easier analysis and comparison.



The screenshot shows a terminal window with progress bars for loading a dataset. Below the progress bars is a table with sentiment analysis results for five samples.

	text	true_label	DistilBERT_pred	DistilBERT_score	RoBERTa_pred	RoBERTa_score	BERT_pred	BERT_score
0	The core message is strong, the cast has given...	NEGATIVE	NEGATIVE	0.997454	NEGATIVE	0.386150	NEGATIVE	0.453077
1	This film is about a young man's painful jour...	NEGATIVE	POSITIVE	0.975371	NEGATIVE	0.501295	POSITIVE	0.405105
2	With a title like that, you will be forgiven f...	NEGATIVE	NEGATIVE	0.994927	NEGATIVE	0.636101	NEGATIVE	0.479811
3	I was recently given this film on DVD as a gif...	POSITIVE	POSITIVE	0.998675	POSITIVE	0.855483	NEGATIVE	0.340471
4	I was looking forward to this based on the rev...	NEGATIVE	NEGATIVE	0.999450	NEGATIVE	0.812834	NEGATIVE	0.613582

Models Used

Three pretrained transformer models from Hugging Face were used for sentiment classification:

1. DistilBERT

Model ID:

`distilbert-base-uncased-finetuned-sst-2-english`

A lightweight and fast version of BERT optimized for sentiment analysis.

2. RoBERTa

Model ID:

`cardiffnlp/twitter-roberta-base-sentiment`

An improved BERT-based model with strong contextual understanding and high precision.

3. Multilingual BERT

Model ID: `nlptown/bert-base-multilingual-uncased-sentiment`

A multilingual sentiment model that predicts sentiment using star ratings, which were converted into POSITIVE and NEGATIVE labels.

Methodology

The project was implemented using Python in Google Colab with libraries such as pandas, transformers, scikit-learn, and Streamlit.

The workflow included:

1. Loading and preprocessing the IMDb dataset
2. Running sentiment prediction using Hugging Face pipelines
3. Standardizing prediction labels
4. Evaluating the models using Accuracy, Precision, Recall, and F1-score
5. Saving the results into Excel files
6. Developing a Streamlit application for interactive prediction

Results

The evaluation results showed that DistilBERT achieved the best overall performance with the highest Accuracy and F1-score.

...	Model	Accuracy	Precision	Recall	F1-score
0	DistilBERT	0.810000	0.825200	0.786700	0.805500
1	RoBERTa	0.766700	0.892200	0.606700	0.722200
2	BERT	0.763333	0.862385	0.626667	0.725869

RoBERTa achieved the highest Precision score, while multilingual BERT produced acceptable results with slightly lower overall performance.

Overall, the experiment demonstrated that pretrained transformer models are highly effective for sentiment analysis tasks. DistilBERT provided the best balance between accuracy, speed, and efficiency.

Demo Screenshots

The following screenshots demonstrate the functionality of the Streamlit sentiment classification application using different input sentences and transformer models.

Sentence 1

- DistilBERT Result
- RoBERTa Result
- BERT Result

Sentiment Classification System

Classify text sentiment using three pretrained Hugging Face models.

Enter a sentence:

I absolutely loved this movie! The acting was brilliant and the story was amazing.

Choose a model:

DistilBERT

Predict

Prediction Result

Predicted Label: POSITIVE

Confidence Score: 99.99%

Sentiment Classification System

Classify text sentiment using three pretrained Hugging Face models.

Enter a sentence:

I absolutely loved this movie! The acting was brilliant and the story was amazing.

Choose a model:

RoBERTa

Predict

Prediction Result

Predicted Label: POSITIVE

Confidence Score: 99.23%

Sentiment Classification System

Classify text sentiment using three pretrained Hugging Face models.

Enter a sentence:

I absolutely loved this movie! The acting was brilliant and the story was amazing.

Choose a model:

BERT

Predict

Prediction Result

Predicted Label: POSITIVE

Confidence Score: 94.84%

Sentence 2

- DistilBERT Result
- RoBERTa Result
- BERT Result

Sentiment Classification System

Classify text sentiment using three pretrained Hugging Face models.

Enter a sentence:

This product is terrible. It broke after one day and the customer service was useless.

Choose a model:

DistilBERT

Predict

Prediction Result

Predicted Label: NEGATIVE

Confidence Score: 99.98%

Sentiment Classification System

Classify text sentiment using three pretrained Hugging Face models.

Enter a sentence:

This product is terrible. It broke after one day and the customer service was useless.

Choose a model:

RoBERTa

Predict

Prediction Result

Predicted Label: NEGATIVE

Confidence Score: 98.35%

Sentiment Classification System

Classify text sentiment using three pretrained Hugging Face models.

Enter a sentence:

This product is terrible. It broke after one day and the customer service was useless.

Choose a model:

BERT

Predict

Prediction Result

Predicted Label: NEGATIVE

Confidence Score: 96.45%

Sentence 3

- DistilBERT Result
- RoBERTa Result
- BERT Result

Sentiment Classification System

Classify text sentiment using three pretrained Hugging Face models.

Enter a sentence:

The film had stunning visuals, but the plot made absolutely no sense whatsoever.

Choose a model:

DistilBERT

Predict

Prediction Result

Predicted Label: NEGATIVE

Confidence Score: 99.88%

Sentiment Classification System

Classify text sentiment using three pretrained Hugging Face models.

Enter a sentence:

The film had stunning visuals, but the plot made absolutely no sense whatsoever.

Choose a model:

RoBERTa

Predict

Prediction Result

Predicted Label: NEGATIVE

Confidence Score: 76.20%

Sentiment Classification System

Classify text sentiment using three pretrained Hugging Face models.

Enter a sentence:

The film had stunning visuals, but the plot made absolutely no sense whatsoever.

Choose a model:

BERT

Predict

Prediction Result

Predicted Label: NEGATIVE

Confidence Score: 52.16%

Sentence 4

- DistilBERT Result
- RoBERTa Result
- BERT Result

Sentiment Classification System

Classify text sentiment using three pretrained Hugging Face models.

Enter a sentence:

The film had stunning visuals, but the plot made absolutely no sense whatsoever.

Choose a model:

DistilBERT

Predict

Prediction Result

Predicted Label: NEGATIVE

Confidence Score: 99.88%

Sentiment Classification System

Classify text sentiment using three pretrained Hugging Face models.

Enter a sentence:

The film had stunning visuals, but the plot made absolutely no sense whatsoever.

Choose a model:

RoBERTa

Predict

Prediction Result

Predicted Label: NEGATIVE

Confidence Score: 76.20%

Sentiment Classification System

Classify text sentiment using three pretrained Hugging Face models.

Enter a sentence:

The film had stunning visuals, but the plot made absolutely no sense whatsoever.

Choose a model:

BERT

Predict

Prediction Result

Predicted Label: NEGATIVE

Confidence Score: 52.16%

Sentence 5

- DistilBERT Result
- RoBERTa Result
- BERT Result

Sentiment Classification System

Classify text sentiment using three pretrained Hugging Face models.

Enter a sentence:

It's not the worst thing I've ever tried, but I expected much more for the price.

Choose a model:

DistilBERT

Predict

Prediction Result

Predicted Label: NEGATIVE

Confidence Score: 98.79%

Sentiment Classification System

Classify text sentiment using three pretrained Hugging Face models.

Enter a sentence:

It's not the worst thing I've ever tried, but I expected much more for the price.

Choose a model:

RoBERTa

Predict

Prediction Result

Predicted Label: NEGATIVE

Confidence Score: 46.20%

Sentiment Classification System

Classify text sentiment using three pretrained Hugging Face models.

Enter a sentence:

It's not the worst thing I've ever tried, but I expected much more for the price.

Choose a model:

BERT

Predict

Prediction Result

Predicted Label: NEGATIVE

Confidence Score: 68.49%

Conclusion

In this project, three pretrained transformer models were used for sentiment analysis on IMDb movie reviews. The models were evaluated using Accuracy, Precision, Recall, and F1-score metrics.

The results showed that DistilBERT achieved the best overall performance, while RoBERTa achieved the highest Precision score. BERT also produced competitive results with good classification capability.

A Streamlit web application was successfully developed to allow users to test sentiment prediction interactively using the three models.

This project demonstrated the effectiveness of transformer-based NLP models in sentiment classification tasks.

References

1- Hugging Face Transformers Documentation:

<https://huggingface.co/docs/transformers>

2- IMDb Dataset:

<https://huggingface.co/datasets/imdb>

3- Streamlit Documentation:

<https://docs.streamlit.io/>

4- Scikit-learn Documentation:

<https://scikit-learn.org/>